Industrial Policy in the Mexican Automotive Sector:

An Analysis of Sustainable Development Programs for Guanajuato

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Introduction

In recent years, the global landscape has been marked by social, economic, and political challenges such as the COVID-19 pandemic, climate change, and supply chain disruptions (Carrillo et al., 2022). These phenomena have contributed to a resurgence of discussions about the role of the State in the economy, in which industrial policy has emerged as an instrument to "ensure that economic growth is green and inclusive, and makes us resilient to environmental, economic, and geopolitical risks," according to statements from the Group of Seven (G7) (Sandoval, 2022).

The intersection of global value chains, industrial policy, and sustainability has become a key topic in the academic literature and debates within economic science. In the Mexican automotive sector, industrial policies have incentivized technology transfer and supplier upgrading as elements to promote competitiveness and growth (Sandoval, 2019; Álvarez Castañón, 2021; Guzman Anaya, 2023). As the world's sixth-largest vehicle producer, Mexico hosts assembly and manufacturing plants of the leading automobile manufacturers, whose activities have underscored the importance of linking industrial upgrading with sustainability

objectives: reducing carbon emissions, promoting energy efficiency, and fostering development (Falck-Reyes, 2023; Sandoval et al., 2024).

Achieving this "green industrial policy" raises new productive linkages and systemic transformations with far-reaching effects on the positioning of automotive companies (UNIDO, 2022, p. 17). As the industry undergoes technological transformations—such as the shift toward electric vehicles (EVs) and the integration of advanced manufacturing processes—some studies have examined the initiatives, regulatory landscape, and government policies guiding the trajectory of this industry in developing countries, including Mexico (Zhao et al., 2019; Altenburg et al., 2021; Asif et al., 2021; Meléndez, 2022; Guzman Anaya, 2023).

Although industrial policy programs have adopted specific criteria for their implementation and evaluation, research on industrial policy is still seeking greater consensus regarding policy models and their medium- and long-term effects (Fuentes & Pipkin, 2018). In Guanajuato, a set of logistical and commercial advantages has contributed to the rise of the automotive industry, while also highlighting the persistence of sustainability challenges (Sandoval et al., 2024).

In this chapter we analyze the sustainability objectives of the Guanajuato 2040 State Development Plan (*Plan Estatal de Desarrollo Guanajuato 2040*, PEDG)—"the instrument through which different sectors of society agree on the objectives, strategies, and projects for the development of the state with a long-term vision"—whose criteria incorporate, for the automotive industry: effectiveness in attracting foreign investment, promotion of research, local development, job creation, and in particular, the integration of sustainable practices in the production chain (PEDG, 2018). The chapter is organized as follows: the first section addresses theoretical references on industrial policy and sustainability; the second situates Mexican industrial policy in the automotive sector in Guanajuato; the third analyzes the sustainability objectives of the Guanajuato 2040 State Development Plan as linked to the automotive industry. Finally, we reflect on industrial policies regarding sustainability in the

Mexican automotive industry, elaborating a discussion of the effects of technological change and global value chains.

By analyzing the specific sustainability objectives and initiatives of the State Development Plan, this study provides a framework for situating green industrial policy in the Mexican automotive sector, focusing on the implications of technological change and its effects on productive chains. This framework seeks to lay the groundwork for future research on the trajectory of the Mexican automotive industry and its effects on sustainable development objectives, with attention to regional-level reform efforts.

1 Industrial Policy and Sustainability

Chang (1994, p. 66) defines the study of industrial policy as being "directed at particular industries (and firms as their components) to achieve outcomes that the State perceives as efficient for the economy as a whole." The functions of industrial policy can be related to the matching of firms within a technological system, the behavior of economic agents, and the creation of institutional architectures (such as educational, legal, or financial ones) (Chang et al., 2013). Its application involves interdependent processes of learning and production, in which effective coordination between the State and firms is a critical factor for success (Cohen & Levinthal, 1989; Lundvall & Borrás, 2005).

Debates exist about the uses of industrial policy and technical questions regarding the measurement of its effects (Zhao et al., 2019; Juhász et al., 2022). Traditionally, industrial policy has aimed to improve the productivity and competitiveness of firms, implying a "trickle-down" approach in its effects on economic development (Matthews, 2020; Ferranini et al., 2021). In recent years, debates on industrial policy have explored measures for managing structural change and promoting environmental sustainability, highlighting alternative visions of development (Altenburg & Assmann, 2017). These policies seek to leverage changes in global

value chains, the international division of labor (Pipkin & Fuentes, 2017), and technological changes such as automation and the use of new energy sources (Bailey et al., 2019).

This "green industrial policy" has been promoted by national governments and international organizations (such as UNIDO, UNCTAD, and UNDP), which have encouraged the use of low-carbon and more energy-efficient technologies in order to decouple industrial development from resource depletion, pollution, and waste production (Mathews, 2020; UNIDO, 2023). These objectives are embedded in the United Nations 2030 Agenda for Sustainable Development (UN, 2015), which adopts the so-called "triple bottom line approach to human well-being" (Sachs, 2012, p. 2206) by balancing three dimensions of sustainable development: economic, social, and environmental.

The UNIDO Industrial Development Report 2022 (UNIDO, 2023, p. 24) notes that, over the long term, industrial greening will affect the balance of competitive advantages of companies in established industries and the countries that host them. According to the Report, a variety of economic policies—which constitute *de facto* industrial policies insofar as they aim to change the structure of the economy (Table **??**)—must play a central role in post-COVID-19 pandemic recovery programs through the adoption of sustainability standards, collaboration with private entities for project financing, the introduction of clean technologies, and policies to stimulate demand for low-carbon technologies.

In the automotive sector, these technological changes are led by the shift from internal combustion engine vehicles to electric vehicles, whose development is reshaping the composition of global value chains, production processes, labor costs, and the characteristics of employment (Zhao et al., 2019; Pichler et al., 2021). The transition toward this paradigm exerts pressure on governments to reposition their automotive regions, accentuated by the emerging combination of technological systems, components, and automotive parts that make up global production chains (Asif et al., 2021; Goracinova et al., 2022).

In the case of emerging countries, the challenges are compounded by the lack of cross-cutting coordination between industrial policies and their relationship with science and technology policy instruments (Meckling & Nahm, 2019; Casalet & Stezano, 2020). Given the technological content of AI-driven transformation challenges and the lag in this area, the Mexican automotive industry faces significant threats, primarily from the signing of the United States–Mexico–Canada Agreement (USMCA), the disruption of supply chains following the COVID-19 pandemic, and its effects on parts suppliers and workers (Covarrubias, 2021; Carrillo et al., 2022). The Original Equipment Manufacturers (OEMs) present in Mexico are adopting new strategies regarding vehicle production and parts supply (Meléndez, 2022), requiring the design of policies that promote the upgrading and training of firms to implement new processes and technologies (Guzmán Anaya, 2023).

This transition has led automotive companies to rely on partners such as SMEs and public research organizations (Meléndez, 2022). While this context represents an opportunity for domestic firms, their transition has been documented as slow (Casalet & Stezano, 2020) relative to the opening of opportunities, which may lead to a deepening of the division of labor in value chains—keeping Mexico in the manufacturing and assembly segments of auto parts and vehicles—and making it difficult to advance toward meaningful design tasks and the production of high-technology components.

2 Industrial Policy in the Mexican Automotive Sector: The Case of Guanajuato

Following the end of import-substitution industrialization, economic policy in Mexico was characterized by a gradual abandonment of industrial policy as an explicit foundation for development. These measures were characterized by driving an industry based on trade openness and export promotion (Calderón Villarreal & Hernández, 2016), as seen in the

National Program of Industrial Development and Foreign Trade (PRONAFICE, 1984–1988), the National Program of Industrial Modernization and Foreign Trade (PRONAMICE, 1990–1994), the Program of Industrial Policy and Foreign Trade (PROPICE, 1996–2000), and the Business Development Program (PDE, 2006–2011) (Calderón Villarreal et al., 2017).

Despite this, planning instruments have prevailed in the automotive sector at both the federal and state levels (Vázquez & Orteza, 2023). In 2012, the federal government implemented the "Strategic Program for the Automotive Industry 2012–2020" (PEIA), which integrated policies promoting the terminal industry, the domestic market for new vehicles, and the diversification of investments—amid criticism that the automotive industry should generate greater linkages with the rest of the economy rather than merely responding to the needs of foreign capital (Ruiz Durán, 2016; Crossa & Ebner, 2020).

The absence of more comprehensive federal-level strategies has driven policy among state governments. The signing of the North American Free Trade Agreement (NAFTA) resulted in the relocation of the automotive sector from the industrialized regions of the center toward emerging areas in the North and the Bajío, with Guanajuato emerging as a key state (Martínez et al., 2017; Guzman Anaya, 2018). With fiscal, geographical, and infrastructure advantages, Guanajuato is home to Mexico's largest automotive cluster, with seven of the world's leading OEMs present across 23 municipalities (Zárate Negrete & Sánchez Ramos, 2021). Guanajuato is the second state in terms of automotive production value. The automotive industry accounts for more than 50% of the state's gross industrial output and approximately 80% of its exports. The industry comprises more than 400 OEM suppliers and attracts FDI exceeding $20 billion USD (Gobierno del Estado de Guanajuato, 2024).

The industrialization of Guanajuato, driven by the installation of General Motors plants in the 1990s and 2000s, generated supply linkages in auto parts and attracted manufacturers such as Mazda, Honda, Volkswagen, Pirelli, Michelin, GKN Driveline, Schaeffler, Hella, and American Axle (Martínez & Carrillo, 2019). The period also witnessed an expansion

toward more capital- and skills-intensive activities, reflected in the opening of Ford Motor Company and Toyota plants in 2017 and 2019, respectively (Gobierno del Estado de Guanajuato, 2024). This growth strengthened linkages in the supply of auto parts and associated industries (plastics, metals, textiles, and electronics), and led to the diversification of functions, including final assembly, multi-tier suppliers, and specialized activities in design and Research & Development (R&D) (Manzano et al., 2017):

> The growth of educational offerings in Engineering, Manufacturing, and Construction is a reference for the educational relevance in accordance with the current economic environment of the state, since the central region has achieved strong investment attraction with the installation of technology-oriented companies—especially in the automotive sector—which has allowed related supplier companies to require specialized personnel and, consequently, institutions have had to reorient their educational offerings to meet demand. (PEDG, 2018, p. 39)

Industrial policy has been cited as a factor that has driven FDI flows and the growth of the automotive industry in Guanajuato (García Garnica & Martínez-Martínez, 2017). This industrial policy has consisted of: public investment in infrastructure (roads, railways, lighting); the award of scholarships for the training of specialists (design and simulation, hydraulics, robotics, and pneumatics) through the "Bécate" program of the Ministry of Labor and Social Welfare (STPS) and the Federal Training Institute; incentives for hiring workers; tax exemptions; the opening of facilities to assist with compliance with regulations and bureaucratic processes; sponsorship of air tickets for foreign trainers; and financial incentives for plant expansion projects (Martínez & Carrillo, 2017).

Although Guanajuato has been administered by governments of different political parties, all have emphasized the upgrading of automotive companies and the implementation of sectoral policies (Zárate Negrete & Sánchez Ramos, 2021). In the "Guanajuato Siglo XXI (1992–1995)" report, the state government highlighted the need to promote industrialization

and economic diversification, identifying the automotive industry as key to the integration of productive chains. It established the goal of attracting foreign investment to generate employment (Gobierno del Estado de Guanajuato, 1995, p. 1133). Notably, the report emphasized regulatory and environmental regulation support, including the possibility of reimbursing the hiring costs of Mexican firms by the Ministry of Sustainable Economic Development (SDES) (García Garnica & Martínez-Martínez, 2017).

Following the implementation of the Guanajuato Siglo XXI report, the New Planning Law of Guanajuato, enacted in 2000, mandated the incorporation of long-term planning. In 2011, the State Planning Institute was designated as the key coordinator of public policies in the state. Three State Development Plans were implemented in Guanajuato for 2025, 2030, and 2035 (enacted in 2000, 2005, and 2015, respectively) (Manzano et al., 2017).

The update of the State Development Plan 2040 was carried out by the Institute of Planning, Statistics, and Geography of the State of Guanajuato (IPLANEG) and was published in 2018. This process involved the participation of agencies and entities of the public administration, members of the three branches and levels of government, including the State Congress, the Judicial Branch, and federal and municipal public administration authorities (PEDG, 2018). These participants met with representatives of the productive and social sectors to define elements such as the vision, objectives, strategies, and projects contemplated in the instrument.

The PEDG is organized into sections addressing economic, social, and environmental aspects. An introductory section outlines the plan's goals and objectives. The human and social dimension evaluates programs for education, healthcare, and well-being. The economic dimension describes policies to foster growth and job creation. The environment section prioritizes a specific area outlining ecological goals and conservation efforts. The public administration dimension focuses on transport, services, and public goods. In the final section, regional development, the particular implementation and monitoring mecha-

nisms for the state are described. As will be discussed below, the PEDG 2040 interprets the results of industrial policies in the region and their role in shaping the automotive trajectory as part of its sustainable development objectives (Crossa & Ebner, 2020; Meléndez, 2022).

3 Sustainability Outcomes in Mexican Industrial Policy: An Analysis of the Guanajuato 2040 State Development Plan

The benefits of the automotive industry in Guanajuato are reflected in its contribution to Gross Domestic Product (GDP), inflows of Foreign Direct Investment (FDI), the establishment of foreign-owned companies, job creation, technology transfer and technological learning, capacity building, and the creation of new firms. Its secondary effects in environmental terms, however, reveal a polluting industry that is intensive in the use of natural resources (Sandoval et al., 2024).

Industrial policy in Guanajuato is stipulated in the State Development Plan 2040 (PEDG). The plan allocates funds to infrastructure projects—including transportation, energy, and technology—with the aim of improving connectivity and competitiveness, as well as promoting the adoption of renewable energies and emphasizing economic diversification:

> Guanajuato faces two major challenges: on the one hand, the concentration of exports, which is manifested in a share of automotive assembly companies of 75 percent of total exports; and on the other hand, the stagnation of traditional sectors, with the exception of the agro-food sector. Additionally, 85 percent of exports are destined for the countries of the North American Free Trade Agreement, which is in the process of renegotiation. (PEDG, 2018, p. 72)

Martínez and Carrillo (2019) highlight how, in recent years, industrial policy in Guanajuato

has shifted from a "passive" approach aimed at attracting and favoring FDI, toward an "active" one aimed at strengthening the productive capacities of the industry (Figures 1 and 2).

[Figure 5.1: Projects Supported under Investment Attraction Programs, 2007–2011]

Figure 1: Projects supported under investment attraction programs, 2007–2011. *Source:* Martínez and Carrillo (2019).

[Figure 5.2: Projects Supported under Investment Attraction Programs, 2012–2018]

Figure 2: Projects supported under investment attraction programs, 2012–2018. *Source:* Martínez and Carrillo (2019).

> Boosting current economic engines based on primary production to increase productivity; managing new economic models based on the development of the knowledge economy; leveraging potential value-added linkages toward the consumer and the efficient development of supply chains; promoting entrepreneurship and business development of new firms and youth; promoting commercial and economic integration by leveraging trade agreements and treaties; and, finally, consolidating the progressive improvement of the state's industrial and logistics infrastructure. (PEDG, 2018, p. 58)

Table **??** details the industrial policy action plans of the 2035 Plan, classifying lines of action as passive, horizontal, and indirect following Manzano et al. (2018). Passive actions include the elimination of bureaucratic procedures, while horizontal actions indicate broad support across all sectors. In addition, actions are indirect in that they strengthen technological assistance and infrastructure:

> The Technological Parks and Innovation System is composed of seven parks whose objective is to generate a favorable ecosystem for innovation through technology transfer services, creation of technology-based firms, hosting, commercial validation, as well as the appropriation and protection of intellectual property. (PEDG,

2018, pp. 53–54)

During the COVID-19 pandemic, automotive companies saw their complex global supply chains disrupted, particularly in semiconductor production. As a result, automakers are attempting to exercise more significant control over their value chains through nearshoring or reshoring processes (Carrillo et al., 2022; Falck-Reyes, 2023). Traditionally, automotive companies subcontracted most of the manufacturing process in order to focus on design, supplier management, marketing, and parts assembly activities. The recent trend favors in-house production with significant investments in the acquisition of raw materials such as lithium, graphite, cobalt, and nickel (Guzman Anaya, 2018).

This "producer-driven" structure—where manufacturers in developed countries organize production through international supplier networks (Carrillo et al., 2012; Fuentes & Pipkin, 2018)—imposes standards and requirements as conditions of technological integration whose effects on the energy transition in Guanajuato have been limited. For the state government, repositioning traditional automotive regions has led to the implementation of planning instruments to adopt these technologies (Bailey et al., 2019).

The transition toward electric vehicle (EV) manufacturing involves economic, technological, infrastructure, and policy factors. Access to the global supply chain of components and cutting-edge technologies, particularly in battery manufacturing, is limited. The regulatory and normative environment requires the implementation of supportive measures, such as tax incentives, subsidies, and emissions regulations, as well as the creation of a skilled workforce in areas such as battery manufacturing and software development (Olguín et al., 2023).

Plans are needed to accelerate the introduction of electric vehicles over the next decade and expand battery production, especially given the restructuring of value chains damaged during the COVID-19 pandemic. This has spurred offshoring and in-house manufacturing among major automobile assemblers, generating the need to discuss the effects of this transition on

SME suppliers in Mexico (Falck-Reyes, 2023):

> Guanajuato has a diverse system of consolidated economic clusters articulated with the logistics, production, and local supply sectors. It is necessary to increase the productivity of small and medium-sized enterprises and local supply chains through the professionalization of the workforce, the business base, and the incorporation of innovation into productive processes in order to make them technologically competitive. (PEDG, 2018, p. 53)

The PEDG 2040, drafted in 2018, lacks provisions for the development of electric vehicles. The automotive industry has experienced rapid advances in electric vehicles since then: advances in battery technology and charging infrastructure, changes in consumer preferences, environmental impact, and the success of developing countries such as China make it necessary to reconsider this omission (Zhao et al., 2019; Altenburg et al., 2022; Asif et al., 2022). In this regard, implementing policies in which foreign and domestic multinational companies, as well as local governments and supplier associations, participate in a coordinated manner is desirable and feasible (Álvarez Castañón, 2021; Basave et al., 2023) in order to encourage the establishment of automotive facilities and promote processes of technological collaboration on sustainability.

Discussion

Following trade liberalization, Mexican industrial policy has consisted of initiatives centered on attracting FDI and promoting exports, without formulating diversified strategies in accordance with the heterogeneity of industrial regions in the country. This has led to a weak incorporation of domestic firms into tasks of increasing added value (Sandoval, 2019). The evolution of industrial policy in Guanajuato—from a passive approach of attracting foreign investment toward a more active one focused on strengthening productive capacities—reflects

the resilience of Mexican regions in the face of emerging opportunities in global value chains, and thereby the demand for adaptive, region-centered industrial policies coherent with the macroeconomic scenario (Crossa & Ebner, 2020; Basave et al., 2023).

Despite the economic benefits generated by automotive industry activity in Guanajuato, significant challenges are observed in sustainability policies (Sandoval et al., 2024). The industrial policies reflected in development plans have focused on establishing positive linkages for foreign companies, technical training, and the integration of local suppliers into upgrading processes. The sustainability measures outlined in the PEDG 2040 focus on economic growth, competitiveness, and innovation, often without addressing the unsustainable practices of the automotive industry.

As the shift toward an electric vehicle paradigm gains momentum, according to Goracinova et al. (2022), the semi-periphery risks being relegated to a more limited assembly role, due to the substantially smaller number of parts and components that make up electric vehicles and, in Mexico's case, the lack of an institutional energy infrastructure in line with international market trends.

The lack of provisions and industrial policy measures toward electric vehicle manufacturing limits the effectiveness of the Plan in addressing sustainability challenges and opportunities in Guanajuato's automotive sector. At the international level, OEMs demanding greater in-house production of batteries, components, and software will have a displacement effect on suppliers that fail to meet these requirements (Asif et al., 2022; Falck-Reyes, 2023; Guzman Anaya, 2023). In this context, the role of government in fostering sectoral innovation is crucial for shaping markets and upgrading in global value chains.

The sustainability objectives of the Plan illustrate the need for an industrial policy grounded in responding to social challenges such as climate change, demographic transitions, health, well-being, and sustainability—systemic problems that require knowledge from different per-

spectives and institutional logics (Mazzucato, 2018). The critical discussion of industrial policies is fundamental to advancing in this direction. In emerging regions such as Guanajuato, the new industrial policy must focus on developing capacity, resilience, and inclusion, while providing public goods such as robust infrastructure and clean energy.

The challenge lies in encouraging investment decisions that promote development paradigms and orient the market toward more sustainable outcomes. The regional and fragmented character of previous strategies highlights the need for greater integration in national-level development plans. The design of industrial policy instruments must drive structural change toward sustainable development, emphasizing the resilience of productive chains as a prerequisite to ensuring "a reliable, diverse, and codependent global supply, consistent with the values of free trade" (Sandoval, 2022).

References

Altenburg, T., Corrocher, N., & Malerba, F. (2022). China's leapfrogging in electromobility: A story of green transformation driving catch-up and competitive advantage. *Technological Forecasting and Social Change, 183*, 121914.

Altenburg, T., & Assmann, C. (Eds.) (2017). *Green industrial policy: Concept, policies, country experiences.* UN Environment; German Development Institute (DIE).

Álvarez Castañón, L. (2021). La transferencia de tecnología 4.0: complejidades, condiciones y contradicciones en el ecosistema de innovación. In S. González López, R. Hernández Mar, & R. Rozga Luter (Eds.), *Digitalización de la guerra, lo humano, el arte y los espacios urbanos y productivos.* Universidad Autónoma Metropolitana.

Asif, M., Jajja, M. S. S., & Searcy, C. (2021). A review of literature on the antecedents of electric vehicles promotion: Lessons for value chains in developing countries. *IEEE*

Transactions on Engineering Management.

Bailey, D., Glasmeier, A., Tomlinson, P. R., & Tyler, P. (2019). Industrial policy: New technologies and transformative innovation policies? *Cambridge Journal of Regions, Economy and Society, 12*(2), 169–177.

Basave, J., Sandoval, S., & Montiel, P. (2023). De la atracción de la inversión extranjera directa a la incorporación de valor nacional: Experiencias de cadenas globales de valor en México. In J. Basave, *Balance de la política económica en México* (pp. 105–142). Instituto de Investigaciones Económicas, Universidad Nacional Autónoma de México.

Calderón Villarreal, C., & Hernández, L. (2016). Cambio estructural y desindustrialización en México. *Panorama Económico, 11*(23), 153–190.

Calderón Villarreal, C., Vázquez, B. I., & López Valdez, L. I. (2019). Evaluación de la política industrial durante el período de apertura económica en México. *Nóesis. Revista de ciencias sociales, 28*(55), 162–184.

Carrillo, J. (2012). Multinacionales en México y su participación en la cadena de valor. In J. Carrillo (Ed.), *Dilemas de la innovación en México.* El Colegio de la Frontera Norte.

Carrillo, J., Martínez, A., López, O., & Díaz, M. (2023). *Impact of automation and digitalization on employment.* International Labor Organization; European Commission. https://www.ilo.org/wcmsp5/groups/public/---ed_emp/documents/publication/wcms_877324.pdf

Carrillo, J., Vallejo, B., & Gomis, R. (2022). COVID-19 and industrial resilience in the Global South: A case study on the auto parts sector in Mexico. *International Journal of Automotive Technology and Management, 22*(1), 82–105.

Casalet, M., & Stezano, F. (2020). Risks and opportunities for the progress of digitalization

in Mexico. *Economics of Innovation and New Technology, 29*(7), 689–704.

Chang, H. J., & Amsden, A. H. (1994). *The political economy of industrial policy* (p. 112). Macmillan.

Chang, H. J., Andreoni, A., & Kuan, M. L. (2013). *International industrial policy experiences and the lessons for the UK.*

Cohen, W. M., & Levinthal, D. A. (1989). Innovation and learning: The two faces of R&D. *The Economic Journal, 99*(397), 569–596.

Covarrubias, A. (2021). El T-MEC y la tercera generación de arreglos laborales: Escenarios probables para el trabajo y la industria regional. *Norteamérica, 16*(1), 147–190. https://doi.org/10.22201/cisan.24487228e.2021.1.470

Crossa, M., & Ebner, N. (2020). Automotive global value chains in Mexico: A mirage of development? *Third World Quarterly, 41*(7), 1218–1239.

Durán, C. R. (2016). Desarrollo y estructura de la industria automotriz en México. *Friedrich Ebert Stiftung*, (6).

Falck-Reyes, M. (2023). Japanese cooperation and suppliers in Mexico's automotive sector: New challenges from USMCA and nearshoring. In *Japanese Cooperation and Supporting Industry in Mexico's Automotive Sector: USMCA, Covid-19 Disruptions, and Electric Vehicle Production* (pp. 7–30). Springer Nature Singapore.

Ferrannini, A., Barbieri, E., Biggeri, M., & Di Tommaso, M. R. (2021). Industrial policy for sustainable human development in the post-Covid19 era. *World Development, 137*, 105215.

Fuentes, A., & Pipkin, S. (2018). Neither synthesis nor rivalry: Complementary policy models and technological learning in the Mexican and Brazilian petroleum and automo-

tive industries. *Business and Politics, 21*(1), 113–144. https://doi.org/10.1017/bap.2018.23

García Garnica, A., & Martínez-Martínez, A. (2018). Trends of automotive industry FDI in Guanajuato, Mexico. *Análisis económico, 33*(84), 99–122.

Gobierno del Estado de Guanajuato (1995). *Guanajuato Siglo XXI*. Gobierno del Estado.

Gobierno del Estado de Guanajuato (2018). *Plan Estatal de Desarrollo 2040*. Gobierno del Estado.

Gobierno del Estado de Guanajuato (2024). Ecosistema de negocios automotriz. https://invierte.guanajuato.gob.mx/automotive/

Guzman-Anaya, L. (2018). The impact of Japanese training programs on local automotive suppliers' productivity in Mexico's State of Guanajuato. In *Japanese Direct Investment in Mexico's Transport Equipment Sector: Macro Impact and Local Responses* (pp. 103–133).

Guzman-Anaya, L. (2023). The role of Japanese cooperation in the transition to electric vehicle production in Mexico. In *Japanese Cooperation and Supporting Industry in Mexico's Automotive Sector: USMCA, Covid-19 Disruptions, and Electric Vehicle Production* (pp. 31–52). Springer Nature Singapore.

Juhász, R., Lane, N., Oehlsen, E., & Pérez, V. C. (2022). *The who, what, when, and how of industrial policy: A text-based approach.*

Lundvall, B. Å., & Borrás, S. (2005). Science, technology, and innovation policy. In *The Oxford Handbook of Innovation* (pp. 599–631). Oxford University Press.

Manzano, F. J. M., Atrizco, V. H. G., & Villavicencio, D. (2018). La política de CTI e industrial de Guanajuato: Un análisis desde la especialización inteligente. *Teuken Bidikay: Re-*

vista Latinoamericana de Investigación en Organizaciones, Ambiente y Sociedad, 9(12), 95–118.

Manzano, F. J. M. (2014). *La especialización inteligente: implicaciones para la política de ciencia, tecnología e innovación de Guanajuato* [Master's thesis]. Universidad Autónoma Metropolitana.

Martínez, A. M., Navarro, G. S., & Garnica, A. G. (2017). Especialización productiva y análisis relacional: El caso de la industria automotriz en Guanajuato. *Frontera Norte, 29*(58), 121–140.

Martínez, A. M., & Carrillo, J. (2019). Evolución de la política industrial en regiones emergentes: El caso de la industria automotriz en Guanajuato, México. *Estudios Sociales. Revista de Alimentación Contemporánea y Desarrollo Regional, 29*(54), 2–30. https://doi.org/10.24836/es.v29i54.757

Martínez, A. M., & Carrillo, J. (2017). *¿Hay política industrial en Guanajuato? Análisis de la industria automotriz.* Clave Editorial.

Mathews, J. A. (2020). Greening industrial policy. In A. Oqubay, C. Cramer, H.-J. Chang, & E. R. Kozul-Wright (Eds.), *The Oxford Handbook of Industrial Policy.* Oxford University Press.

Mazzucato, M. (2018). Mission-oriented innovation policies: Challenges and opportunities. *Industrial and Corporate Change, 27*(5). https://doi.org/10.1093/icc/dty034

Meckling, J., & Nahm, J. (2019). The politics of technology bans: Industrial policy competition and green goals for the auto industry. *Energy Policy, 126*, 470–479.

Meléndez, S. (2022). La inversión extranjera directa en la industria automotriz en México y su efecto en las Mipymes. In S. Ortiz (Ed.), *Inversión extranjera directa y desarrollo*

económico en América Latina: una lectura crítica. Facultad de Economía, Universidad Nacional Autónoma de México.

Olguín, F. P., Iskakov, G., & Kendall, A. (2023). US–Mexico second-hand electric vehicle trade: Battery circularity and end-of-life policy implications. *Transportation Research Part D: Transport and Environment, 125*, 103934.

Pichler, M., Krenmayr, N., Schneider, E., & Brand, U. (2021). EU industrial policy: Between modernization and transformation of the automotive industry. *Environmental Innovation and Societal Transitions, 38*, 140–152.

Pipkin, S., & Fuentes, A. (2017). Spurred to upgrade: A review of triggers and consequences of industrial upgrading in the global value chain literature. *World Development, 98*, 536–554.

Sachs, J. D. (2012). From millennium development goals to sustainable development goals. *The Lancet, 379*(9832), 2206–2211.

UN (2015). *Transforming our world: The 2030 Agenda for Sustainable Development.* Resolution adopted by the General Assembly (A/RES/70/1), 25 September 2015.

Sandoval, S., Sánchez, M. D., & López, V. C. (2024). El consumo de agua en la cadena de suministro de la industria automotriz: El caso de Guanajuato, México. *Intersticios Sociales*, (28).

Sandoval, S. (2019). Upgrading y competencia: Reflexiones para firmas y países en desarrollo. *Economía UNAM, 16*(48), 213–248.

Sandoval, S. (2022). México frente al cambio de paradigma: Del Consenso de Washington al Consenso de Cornwall (pp. 329–348).

Vázquez, R. M. C., & Ortiza, G. A. C. (2023). Impacto de la industria automotriz en el

desarrollo regional en México. *Problemas del Desarrollo. Revista Latinoamericana de Economía, 54*(213). https://doi.org/10.22201/iiec.20078951e.2023.213.69963

Zárate Negrete, L. E., & Sánchez Ramos, M. E. (2021). Incertidumbre en la gestión de capital humano ante la contingencia COVID-19 en la industria automotriz del estado de Guanajuato, México. *RIDE. Revista Iberoamericana para la Investigación y el Desarrollo Educativo, 11*(22).

Zhao, Q., Li, Z., Zhao, Z., & Ma, J. (2019). Industrial policy and innovation capability of strategic emerging industries: Empirical evidence from Chinese new energy vehicle industry. *Sustainability, 11*(10), 2785. https://doi.org/10.3390/su11102785

Table 1: A Typology of Industrial Policies: Innovation, Science, and Technology

Science Policy	Innovation Policy	Technology Policy
Focus: **Production of scientific knowledge** ***Instruments:*** • **Competitive public research grants** • **Public research institutions (labs, universities, research centers)** • **Tax incentives for firms** • **Higher education** • **Intellectual property rights**	*Focus:* General innovative performance of the economy *Instruments:* • Improving individual skills and learning capacities • Improving organizational performance and learning (ISO 9000, quality control) • Improving access to information • Environmental regulation • Bioethical regulation • Corporate law • Competition regulation • Consumer protection • Social capital for regional development (clusters, industrial districts) • Intellectual benchmarking • Smart, reflexive, democratic foresight	*Focus:* Advancement and commercialization of sectoral technical knowledge *Instruments:* • Public procurement • Public support for strategic actors • Bridge institutions between research and industry • Standardization • Technology forecasting • Industrial sector benchmarking

Source: Own elaboration based on Lundvall and Borrás (2005).

Table 2: Evolution of Industrial Policy Action Plans in Guanajuato

STI Policy		**Industrial Guidelines**	
Period/Phase	**Action Plans**	**Period/Phase**	**Action Plans**
Strengthening of STI; Leveraging science	PCITEG 1998–2020	Education for competitiveness; Innovation and technological development	PED 2035
Capitalizing knowledge for state well-being	PECYT GTO 2030	Business and employment; Infrastructure and logistics	

Source: Manzano et al. (2014).